Analog Biometric Recognition System

Time-Series Signal Processing & Biometric Authentication
Public text edition — institutional email removed

Abstract

This project presents a biometric identification system based on photoplethysmography (PPG) signals using a one-dimensional Convolutional Neural Network (1D CNN). The described workflow includes signal acquisition, preprocessing, segmentation and classification. The academic report records an accuracy of approximately 92–93% and describes stable training behaviour with minor overfitting.

Introduction

PPG is a non-invasive measurement of blood-volume changes. Temporal characteristics such as waveform shape, amplitude variation and rhythm can be used as discriminative features. The project explores CNN-based classification of PPG segments for user identification.

System Overview

The report describes signal acquisition from a sensor connected to a microcontroller, followed by signal filtering, normalization, segmentation, feature extraction and classification. Fixed-length windows of 300 samples are used as CNN inputs.

Data Processing

The report describes noise filtering, baseline correction and z-score standardization before dividing signals into 300-sample segments. The dataset consists of prerecorded PPG signals collected from multiple subjects. The original preprocessing implementation was not recovered and is not included in this public repository.

Model Architecture

The report describes a 1D CNN accepting inputs of shape (300, 1). Its appendix specifies a first Conv1D layer with 32 filters and kernel size 3, max pooling, a second Conv1D layer with 64 filters and kernel size 3, another max-pooling layer, flattening, a dense layer with 128 neurons, dropout of 0.5 and a softmax output layer. The report appendix uses a three-class output and categorical cross-entropy.

Results and Analysis

The report records approximately 92–93% test accuracy. It describes training and validation accuracy curves with a small gap and relatively stable validation behaviour. The exact metric values and original experiment outputs are not available in the recovered source files; the reported accuracy has not been independently reproduced from this repository.

Hardware Signal Observation

The report describes an amplified PPG sensor connected to an ESP32 microcontroller. The acquired waveform showed periodic peaks corresponding to the cardiac cycle and was visualized using the serial plotter. A Doppler-based motion-sensing module was also evaluated for motion-induced signal variations.

Discussion and Conclusion

The report argues that temporal PPG patterns contain discriminative information for biometric identification. It identifies signal conditioning and noise reduction as areas for improvement and notes future work involving larger datasets, stronger robustness and real-time classification.

Recovered implementation caveat

The recovered Python training script differs from the architecture printed in the report. The script uses Dense(2), sparse_categorical_crossentropy, no dropout layer, batch size 16 and validation_split=0.2. The report appendix shows three output classes, categorical_crossentropy, dropout 0.5 and batch size 32. The supplied script is preserved as recovered; the discrepancy must be resolved against original data and experiment files before claiming exact reproduction.

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